

Systems Analysis, Design and Implementation

MIS 6102

Take-Home exam

You have been hired as a consultant, and you are required to apply the principles and techniques of Systems Analysis and Design to a real-world scenario. The client requires that you study their product that was delivered by a software company last year and identify areas for either **Adaptive maintenance** (this involves modifying the software system to keep it usable in a changing environment as a result of new features) or **Perfective maintenance** (this involves modifying the software to improve its performance, maintainability, or usability as requested by the user or identified by the development team. This usually means accommodating new or changed user requirements). You will analyze a given problem, define the solution's requirements and scope, and model key aspects of the proposed modification of system.

Project Overview

Case Study and Challenge Understanding

1. Case Study Analysis: Carefully read and analyze the provided case study description (link to be provided separately).
2. Contextual Research: Supplement the provided information with external research using internet materials to gain a comprehensive, detailed understanding of the business challenge, the existing environment, and the stakeholders involved.
3. Project Description and Scope: Based on your analysis, provide a concise yet thorough description of the proposed project (the system to be developed) and clearly define the scope of what the project will deliver. This section must delineate what is *in scope* and, briefly, what is *out of scope*.

Deliverables and Submission Requirements

Your final submission must comprise two main components: 1. Two PDF documents (i) Project charter and (ii) detailing the analysis and design artifacts, and 2. A video recording of your presentation.

1. A Project Charter
(A Project Charter is a formal, high-level document that officially authorizes the existence of a project and provides the project manager with the authority to apply organizational resources to project activities) See Jack T. Marchewka 2016.
2. System Requirements Specification (SRS) - PDF Document

Develop a comprehensive System Requirements Specification (SRS). You may use the template provided in Lecture 3 or any appropriate industry-standard template (e.g., from IEEE).

The SRS must include the following sections:

- a. Requirements Elicitation

Identify and clearly state two (2) functional requirements and two (2) non-functional requirements for the proposed modified system.

b. Functional Modeling

For the two identified functional requirements, create:

Two Use Case Descriptions (one for each requirement).

Two Use Case Diagrams (UML diagrams illustrating the actors and system boundaries).

c. Detail Modeling

- i. Select one use case scenario, explain it briefly, and draw a sequence diagram based on it.
- ii. Draw a BCE model (or robustness diagram) and convert the BCE model (or robustness diagram) to a collaboration diagram.

d. Detailed Requirements

Develop specific data requirements (e.g., data types, structures, persistence needs) for the system.

Develop detailed performance requirements derived from the non-functional requirements identified in section 1.A (e.g., response time, throughput, capacity).

e. Implementation

Using an Artificial Intelligence (AI) tool of your choice (e.g., Copilot, ChatGPT, etc.), generate a rough code snippet that demonstrates the implementation potential of the use case diagram. Document the tool used and the prompt provided.

3. User Interface (UI) Design and Wireframing

- a. Develop at least two (2) distinct user interfaces (UIs) for the proposed system.
- b. Create wireframes (visual blueprints/skeletons) for these UIs to depict the layout, navigation, and structure of the application screens. *Note: Focus on the structure and low-fidelity design, not high-fidelity visual aesthetics.*

Note: Presentation and Submission

Presentation Requirements

Format	A video recording presenting your work.
Duration	The presentation must not exceed 20 minutes .
Content	Present a project charter, the proposed system description, scope, and the main artifacts from your SRS and UI/wireframe work.

Submission Checklist

1. **2 PDF Files:**

- i. Project charter.

- ii. A professionally formatted System Requirements Specification (SRS) document, incorporating all elements from Deliverable 1 and Deliverable 2.
2. **Video Recording:** The presentation (maximum 20 minutes).

Additional tips:

Review tutorials below.

First, review the tutorial on converting robustness (or BCE) diagrams to collaborations diagrams from here: <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-uml-collaboration-diagram/>

Then, review the tutorial on building a sequence diagram from robustness diagram (or BCE models) from here: <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/robustness-analysis-tutorial/>

This is an open book exam and you can use the books provided and other resources at your disposal.